

Coffee Masterclass: An Experience of Co-Creation with Prompt Engineering and Generative AI for Immersive Environments Development

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Figure 1: Screenshot of Coffee Masterclass experience. All objects in the scene were generated using Generative Artificial Intelligence (GenIA) tools. Similarly, the conceptualization and design stages also include using GenIA tools.

ABSTRACT

This work presents the design and development process of an immersive experience applying a co-creation approach between humans and generative artificial intelligence tools. From the point of view of any user, Coffee Masterclass is an immersive experience that brings anyone to the art and pleasure of preparing specialty coffees. However, the Coffee Masterclass is the result of the inclusion of prompt engineering outputs in each stage of the building process. The co-creation approach is included in all development processes, i.e., from the narrative to the visual content generated through code writing, which has been co-created between the creative team and GenAI. This work tells details of this approach, including how the generative artificial intelligence tools were used in each stage of immersive experience development. This work shows the advantage of involvement in a development team of people with skills in prompt engineering and interaction with Large Language Models. Also, it includes recommendations to other development teams, including generative artificial intelligence tools by future developments.

Index Terms: Computing methodologies—Computer graphics—Graphics systems and interfaces—Mixed/augmented reality

1 INTRODUCTION AND OVERVIEW

“*Prompt Engineering*” is a relatively new term that describes skills to write suitable scripts as input of Generative Artificial Intelligence (GenAI) tools. At the core, these tools use Large Language Models (LLMs), pre-trained deep learning models that allow a

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text prompt to generate a specific output. For this reason, the development and production of interactive multimedia content is evolving due to the introduction of GenAI tools. These changes have allowed GenAI tools as a part of the creativity team, promoting a co-creation approach between humans and GenAI. Threading an immersive experience may also take advantage of this co-creation approach. This work presents the design and development process of an immersive experience applying a co-creation approach between humans and GenAI. Thus, Coffee Masterclass is an immersive experience resulting from inclusion of “*Prompt Engineering*” outputs in each stage of the building process. The result is an immersive experience that brings anyone to the art and pleasure of preparing specialty coffees, “*Coffee Masterclass*”.

This work tells details of the co-creation approach experience of build including how the generative artificial intelligence tools was used in each stage of immersive experience development. The co-creation approach is included in all development processes, i.e., from the narrative to the visual content generated passing through code writing, have been co-created between the creative team and GenAI. Previous works explored this Human-GenAI collaboration as an alternative that allows the breaking of some limitations of Previous works explored this Human-GenAI collaboration as an alternative that allows the breaking of some limitations of designers and developers, with a broad perspective for developing more expressive, accessible, and dynamic experiences [3]. Thus, this co-creation in the development of immersive environments has shown to be a suitable model to be included in the pipeline of different immersive experiences, such as the metaverse application, eXtended Reality experience, among others [3, 1, 2]. However, these tools are emerging, and it is necessary to show in applied cases the advantages, limitations, and recommendations of how to include these in the build process of immersive experiences.

This work presents an applied case that includes a co-creation approach between humans and GenAI in all development processes,

i.e., from the narrative to the visual content generated passing through code writing.

2 COFFEE MASTERCLASS DEVELOPMENT PROCESS

This work is a result from a collaborative process between GenAI tools and a development team with experience in immersive systems development, the Figure 1 shows a screenshot. Currently, a varied ecosystem of GenAI tools covers all stages of development of an immersive system, and the expectation of introducing new tools and their possibilities is exciting.

In the first stages, conceptualization, pre-production, feasibility, and analysis of requirements, The co-creation approach facilitated all processes with a constant dialogue with ChatGPT¹ and Bard². Thus, the title of the immersive experience, the aesthetics concept, the basic requirements, and the narrative are the results of this approach. The main consideration to be taken into account in these early stages of development is that the co-creation must involve a two levels top-down dialogue. At the upper level, the creative team should maintain a traditional dialogue with stakeholders and experts in the field of the experience application, in this case, people interested in the specialty coffee preparation process and expert baristas. From this traditional dialogue, it is important to extract key words and phrases to be used in the inputs of the “Prompt Engineering”. The low-level dialogue includes the co-creation process with GenAI tools, where the goal is to refresh to new ideas that have not been explored with the outputs of the prompts. In the stages of 3D and multimedia production, the co-creation approach used a set of LLMs tools (DALL-E, Midjourney, Stable Diffusion, Leonardo AI, Adobe Firefly, Skybox AI, Krea.AI). However, prompt input always included relevant words/sentences related to narrative and aesthetic concepts. Leonardo AI³ generated images in different visual styles that satisfy the narrative and aesthetic requirements. Skybox AI⁴ facilitated the crafting of images with an equirectangular projection that can be used as Skybox to provide an immersive visual sense of the virtual environment. The images generated with Skybox AI were retouched in Adobe Firefly⁵ to provide a sense of depth and visual atmosphere by combining multiple outputs. Furthermore, to produce aesthetic patterns, offering unique and harmonious color palettes, and carefully selected to align with the graphic concept and application in the user interface. For this, Color Magic⁶ allowed the generation of a color palette commensurate with the graphic concept and aesthetical design. WithPoly⁷ empowered the creation of realistic textures necessary to provide an attractive feeling in the virtual environment.

The co-creation for generating 3D models from a prompt input is possible with emerging LLMs (Kaedim 3D, DreamFusion, 3DFY, Shap-E, Genie), which allows co-creation for generating 3D models from simple prompts. In this case, Genie⁸ and Shap-E⁹ were the tools chosen. With simple prompt inputs, these tools allow generating a set of furniture, nature, or another 3D model to populate the virtual world with good-quality meshes and textures of various assets. Similarly, in video generation, visual sequences allow exploring narratives and visual effects in audiovisual production facilitated by LLMs (Runway, Pika.art, Leonardo AI). Runway¹⁰ allowed generating videos to represent actions and follow the script for step-by-step instructions. It is necessary to remember that the

goal of these co-creation approach stages is to generate 3D and multimedia content faster than traditional production; for these reasons, the more relevant aspect is the constant evaluation and selection of content generated by the LLM. Then, the creative team must review all generated content consideration factors such as color, visual style, context, environment, aesthetics, dimensions, format, and all aspects that contribute to the quality of the result.

The device used to deploy an immersive experience was Display Meta Quest 2, the most known runtime embedded system integrated in a Head Mounted Display. Unity3D was the integrated development tool. This tool was chosen because it is a pioneer in integrating into its curated ecosystem AI-powered solutions, including GenAI tools. It is also recognized for its high-quality graphics, C# programming language support, and compatibility with Meta development SDK and OpenXR standard. In order to accelerate and facilitate the code implementation stage this building process used BlackBox AI¹¹ in the generation of some scripts of C# language. Finally, the testing, evaluation with users and experts, and launch plan may be prepared by a conversational collaboration with ChatGPT and Bard, allowing the generation of dynamic and adaptive strategies providing valuable information to evaluate usability, immersion, and user experience.

3 FINAL REMARKS AND RECOMMENDATIONS

This work showed a case of the application of a co-creation approach between Generative AI tools and a development team to build an immersive application. In the stages of conceptualization, pre-production, and analysis of requirements the co-creation approach allows exploring ideas and concepts to enrich the experience. In the 3D/multimedia production and code implementation stages, the quality of graphics, aesthetics, and narrative is the same as a traditional process, but the process is faster than traditional methods. This co-creation experience between humans and GenAI shows the advantage of a creation team with suitable skills in prompt engineering and interaction with Large Language Models. For this reason, a speculative result of this experience is that skills will be part of the desirable capabilities of the persons involved in creating and developing immersive experiences. Another relevant recommendation is not to drop the outputs of GenAI because any image, model, or sound may be used to enrich the virtual environment. Consider that building a virtual world requires a set of elements, and GenAI outputs are a suitable option to get these elements. However, some tools are emerging, and their outputs must be evaluated in relevance and quality before being included in the environment.

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